Inference of an SIR difference equation model using dynamic survival analysis and Turing.jl - Wave 2 of the Ebola outbreak data

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Introduction

The second wave of the 2018–2020 DRC Ebola epidemic was centred on Butembo, Katwa, Mabalako, and Mandima. This notebook analyses Wave 2 up to 27 May 2019, using 1,104 records and retaining all cases regardless of data completeness.

Wave 2 has the largest share of incomplete records among the three waves: 392 of 1,104 (35.5%) have known onset but no recorded hospitalisation, and 6 have hospitalisation but no known onset. Each group contributes a different likelihood component:

Group	Count	Likelihood contribution
Complete (`e1=1, e2=1`)	706	Multinomial (onset) + Geometric (recovery) + Binomial
Onset-only (`e1=1, e2=0`)	392	Multinomial (onset) + Binomial only
Hosp-only (`e1=0, e2=1`)	6	Convolution over unknown onset + Binomial

Libraries

```
using Turing              # For probabilistic programming
using Distributions       # For probability distributions
using SpecialFunctions    # For lgamma (log gamma function)
using Random              # For random number generation
```

```
using Plots                 # For plotting
using StatsPlots            # For advanced plotting
using MCMCChains            # For handling MCMC output
using Dates                 # For now()
using CSV                   # For reading CSV files
using DataFrames            # For data manipulation
using Statistics            # For statistical functions
```

We set a random seed for reproducibility.

```
Random.seed!(1234);
```

Utility functions

This function converts a rate, `r` over a unit time interval to a proportion.

```
@inline function rate_to_proportion(r, t=1.0)
    if r < 0 || isnan(r)
        return 0.0
    elseif isinf(r)
        return 1.0
    else
        result = 1 - exp(-r * t)
        return isnan(result) ? 0.0 : max(min(result, 1.0), 0.0)
    end
end;
```

This function evaluates `log(exp(a) + exp(b))` by rescaling around `max(a,b)`, avoiding underflow when summing many very small probabilities expressed on the log scale.

```
function logaddexp(a::Real, b::Real)
    m = max(a, b)
    isfinite(m) ? m + log(exp(a - m) + exp(b - m)) : m
end;
```

This function generates a vector of counts from a vector of times and the total number of timesteps in a simulation.

```
function time_counts(times::AbstractVector{<:Integer}, nsteps::Integer)
    counts = zeros(Int, nsteps)
    @inbounds for t in times
        1  t  nsteps && (counts[t] += 1)
    end
    return counts
end;
```

Transitions

We start by defining the deterministic SIR model in discrete time using a system of difference equations. The model tracks the proportion of susceptible (S), infected (I), cumulative infections (C), and new infections per time step (Y).

```
function sir_map!(du, u, p, t)
    (S, I, C, Y) = u
    ( ,  ) = p
    infection = rate_to_proportion( *I)*S     # New infections
    recovery = rate_to_proportion( )*I        # New recoveries
    @inbounds begin
        du[1] = S - infection                  # Update susceptibles
        du[2] = I + infection-recovery         # Update infected
        du[3] = C + infection                  # Update cumulative infections
        du[4] = infection                      # Store new infections
    end
    nothing
end;
```

This function solves a map with the above function signature.

```
function solve_map(f, u0, nsteps, p)
    # Pre-allocate array with correct type
    sol = similar(u0, length(u0), nsteps + 1)
    # Initialize the first column with the initial state
    sol[:, 1] = u0
    # Iterate over the time steps
    @inbounds for t in 2:nsteps+1
        u = @view sol[:, t-1] # Get the current state
        du = @view sol[:, t] # Prepare the next state
        f(du, u, p, t)       # Call the function to update du
    end
```

```
    return sol
end;
```

Alternatively, we could use the `DifferentialEquations.jl` ecosystem (via defining a `DiscreteSystem` and solving using `SimpleFunctionMap` from `SimpleDiffEq.jl`) to simulate the model, although this incurs an overhead in computational cost, in addition to introducing another dependency.

This function simulates the SIR model and returns the solution, the final epidemic size ( ) and the probability distribution of infection times (f).

```
function simulate( ,  ,  , nsteps)
    tspan = (0, nsteps)
    # Preserve type by using zero/one of the promoted type
    T = promote_type(typeof( ), typeof( ), typeof( ))
    zero_T = zero(T)
    one_T = one(T)
    u0 = [one_T -  ,  , zero_T, zero_T] # Initial conditions: S, I, C, Y
    p = [ ,  ]
    sol = solve_map(sir_map!, u0, nsteps, p)
      = sol[3,end]      # Final size (total infected)
    f = Vector{eltype(sol)}(sol[4,2:end])  # Copy to preserve type
    f = abs.(f)         # Ensure non-negative values
    # Handle edge cases for probability vector
    if   <= 0 || any(isnan.(f)) || any(isinf.(f))
        f = fill(one(eltype(f))/nsteps, nsteps)
    else
        f = f/          # Normalise frequencies to sum to 1
    end
    return sol,  , f
end;
```

Data loading

We load the Ebola outbreak data from `EVD_Wave2.csv`.

```
# Load the data
df_all = CSV.read("EVD_Wave2.csv", DataFrame);

# Split into three groups by data availability
df_complete   = df_all[(df_all.event1 .== 1) .& (df_all.event2 .== 1), :]
```

```
df_onset_only = df_all[(df_all.event1 .== 1) .& (df_all.event2 .== 0), :]
df_hosp_only  = df_all[(df_all.event1 .== 0) .& (df_all.event2 .== 1), :]

# Total observed infections for the Binomial final-size term
K_total = nrow(df_all)

# Use all records with a known onset time: complete + onset_only
onset_times_all = vcat(df_complete.onset, df_onset_only.onset)
nsteps = Int(ceil(maximum(onset_times_all))) + 1
ti_onset = max.(1, min.(Int.(round.(onset_times_all)) .+ 1, nsteps))
K_onset  = length(ti_onset)   # used in Multinomial
th_onset = time_counts(ti_onset, nsteps)

# Complete records only: hosp - onset
raw_delta_complete = df_complete.hosp .- df_complete.onset
@assert all(raw_delta_complete .>= 0) "Found complete records with hosp < onset."
same_day_complete = count(<(1), raw_delta_complete)
delta_complete = max.(1, Int.(round.(raw_delta_complete)))

# The "onset" column for e1=0 records contains the hospitalisation time.
T_hosp_only = max.(1, min.(Int.(round.(df_hosp_only.hosp)) .+ 1, nsteps))

# Minimum N: at least as many susceptibles as total observed infections
N_min = K_total
```

Summary of the loaded data:

```
Wave 2 data summary:
  Total records (K_total, Binomial):       1104
  With known onset (K_onset, Multinomial): 1098
    of which complete (e1=1, e2=1):        706
    of which onset-only (e1=1, e2=0):      392
  Hosp-only (e1=0, e2=1, convolution):     6
  nsteps: 90
  Same-day/sub-day complete intervals (<1 day): 89
  Recovery intervals range: 1-36 days
```

Turing Model

The Turing.jl probabilistic model for the model, specifies priors, simulates the difference equations, and defines the likelihood for observed data by defining the distributions of (a) the

total infected (b) the distribution of infection times, (c) the distribution of recovery times and (d) the convolution over unknown onset time for recovery-only records. We place uniform priors on the parameters, as upper and lower bounds are typically easy for the user to define. Here, we take advantage of the discrete nature of the data, by fitting a histogram of the infection times using a multinomial distribution, rather than a series of categorical distributions.

```
@model function dsa_ebola_allrecords(nsteps::Int,
                                     K_total::Int,
                                     K_onset::Int,
                                     th_onset::Vector{Int},
                                     delta_complete::Vector{Int},
                                     T_hosp_only::Vector{Int},
                                     N_min::Int)

    # Priors for model parameters
      ~ Uniform(0.01, 1.0)   # Transmission rate
      ~ Uniform(0.01, 1.0)   # Recovery rate
      ~ Beta(1.0, 500.0)      # Initial infected proportion

    N_float ~ Uniform(Float64(N_min), Float64(N_min + 20000))
    # Convert to integer for Binomial: use floor and clamp to ensure N >= N_min
    # The conversion is done after sampling, so NUTS can sample continuously
    N = Int(floor(max(Float64(N_min), N_float)))

    # Simulate deterministic model
    _,  , f = simulate( ,  ,  , nsteps)

    # Ensure   is valid for binomial distribution
     _valid = clamp( , 1e-6, 1.0 - 1e-6)

    # Ensure f is valid for multinomial distribution
    f_valid = max.(f, 1e-10)           # Add small positive value to avoid zeros
    f_valid = f_valid / sum(f_valid)   # Normalise
    log_f_valid = log.(f_valid)

    # (1) Binomial final-size: all K_total infections
    K_total ~ Binomial(N,  _valid)

    # (2) Multinomial onset histogram: records with known onset (K_onset)
    th_onset ~ Multinomial(K_onset, f_valid)

    # (3) Geometric recovery intervals: complete records only.
    # Build a 1-indexed geometric by shifting Julia's 0-indexed Geometric(p)
```

```
    # so the support matches recovery intervals in days ( = 1,2,...).
       = 1e-8
    p  = clamp(rate_to_proportion( ),  , 1 -  )
    G1 = LocationScale(1, 1, Geometric(p ))
    for i in eachindex(delta_complete)
        delta_complete[i] ~ G1
    end

    # (4) Convolution for hosp-only records: marginalise over unknown onset u.
    #     L_i = Σ_{u=1}^{T_rem-1} f[u] · P(W = T_rem - u), using the same
    #     1-indexed geometric distribution G1 as above.
    for T_rem in T_hosp_only
        u_max = min(T_rem - 1, nsteps)
        log_terms = (log_f_valid[u] + logpdf(G1, T_rem - u) for u in 1:u_max)
        Turing.@addlogprob! foldl(logaddexp, log_terms; init = -Inf)
    end

    return ( = ,  = ,  = , N=N_float)
end;
```

Inference

We now perform inference using the Turing model. We start by setting the number of samples to be used in the sampler and instantiating the model.

```
n_samples = 5000
model = dsa_ebola_allrecords(nsteps, K_total, K_onset,
                             th_onset, delta_complete, T_hosp_only, N_min);
```

Next, we run inference using the No U-Turn Sampler (NUTS).

```
# Warmup
_ = sample(model, NUTS(), 1)
# Run sampler on 4 chains
@time chain = sample(model,
                     NUTS(500, 0.65; max_depth=12, Δ_max=1000.0, init_ =0.2),
                     MCMCThreads(),
                     n_samples,
                     4);
```

We can pass different samplers from `Turing.jl` (e.g. `MH`, `SGLD`, `HMC`), external samplers (e.g. `DynamicHMC.NUTS`, `AdvancedMH.RWMH`) via wrapping with `external_sampler`, and can also pass the model to other packages, e.g. `Pathfinder.jl`.

Results Analysis

Once we have run the sampler, we can get a summary of the parameter estimates, along with the expected number of samples.

```
describe(chain)
```

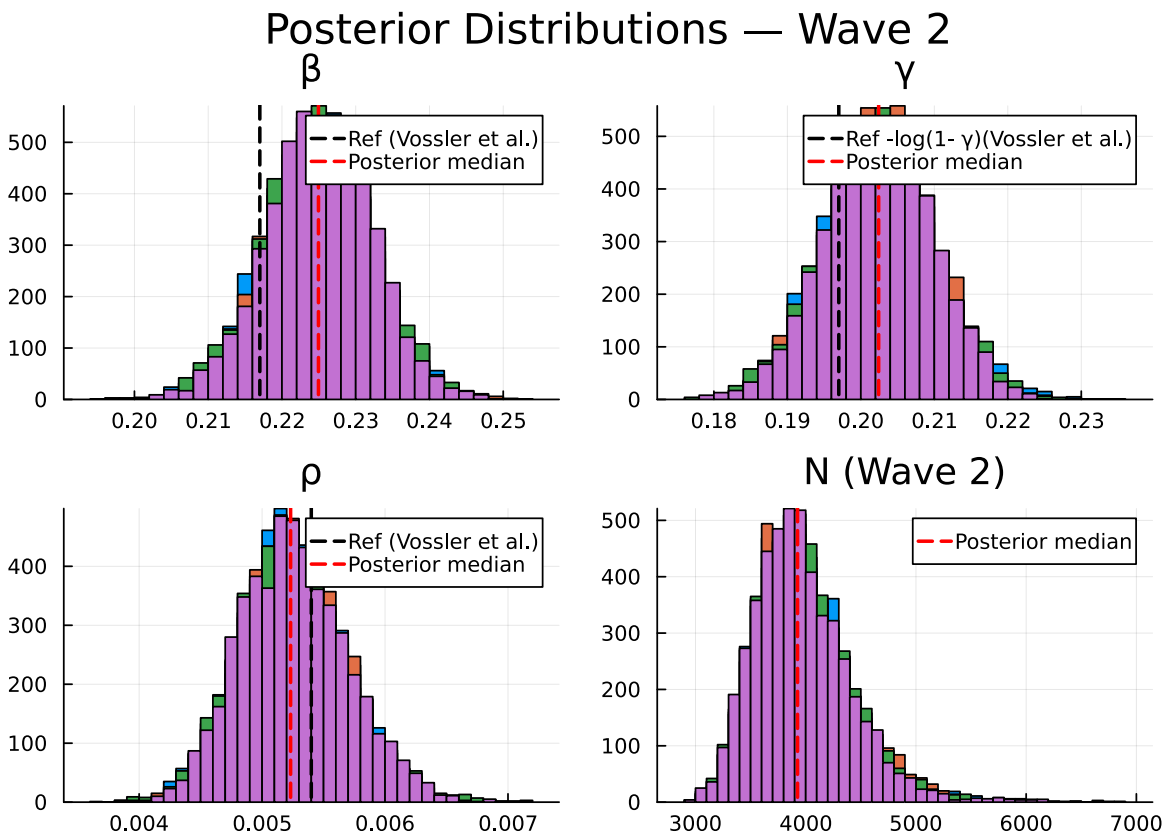
```
Chains MCMC chain (5000×18×4 Array{Float64, 3}):

Iterations        = 501:1:5500
Number of chains  = 4
Samples per chain = 5000
Wall duration     = 883.66 seconds
Compute duration  = 2511.58 seconds
parameters        =  ,  ,  , N_float
internals         = n_steps, is_accept, acceptance_rate, log_density, hamiltonian_energy, ha

Summary Statistics

  parameters        mean        std      mcse    ess_bulk    ess_tail      rha
      Symbol     Float64    Float64   Float64     Float64     Float64   Float6

                  0.2249     0.0073    0.0001   3047.6071   4626.5717    1.000
                  0.2024     0.0075    0.0001   2940.2871   4616.0564    1.000
                  0.0053     0.0004    0.0000   3630.3922   6178.7737    1.000
     N_float   3981.9688   446.6995    6.3888    5460.7332   5474.8743    1.000

                                                                2 columns omitted

Quantiles

  parameters        2.5%       25.0%       50.0%       75.0%       97.5%
      Symbol     Float64     Float64     Float64     Float64     Float64

                  0.2106      0.2200      0.2250      0.2299      0.2391
                  0.1878      0.1973      0.2024      0.2075      0.2173
                  0.0044      0.0050      0.0052      0.0055      0.0062
     N_float   3280.3611   3667.8617   3925.8462   4230.7837   5009.2867
```

NUTS diagnostics:

```
NUTS internals available: [:n_steps, :is_accept, :acceptance_rate, :log_density, :hamiltonian
Divergent transitions: 0 / 20000
Max tree depth reached: 12.0
Step size   (median [min, max]): 0.00186 [0.0013, 0.0209]
```

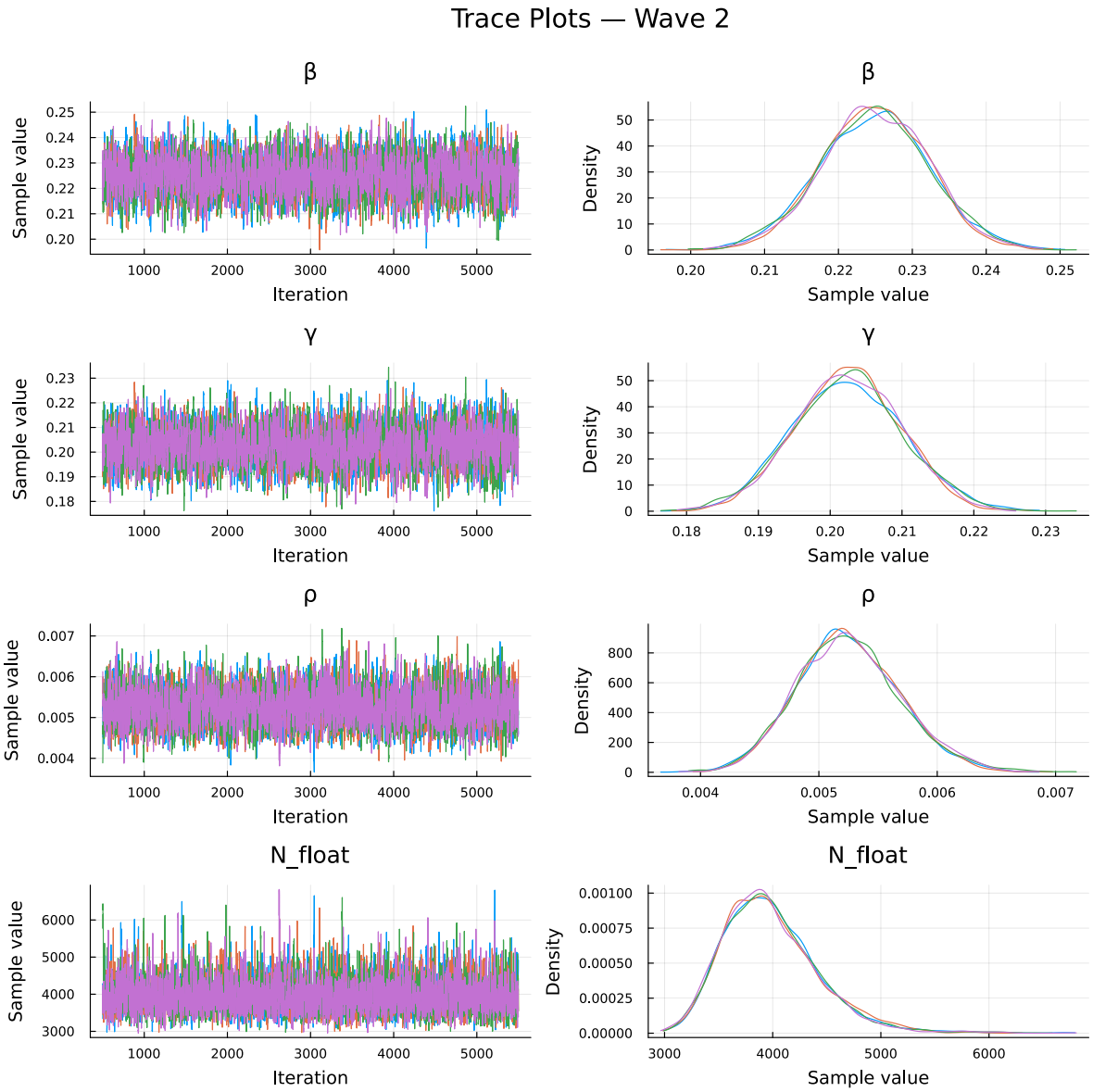
We can also plot posterior distributions for each parameter.

```
 _med = median(chain[:])
 _med = median(chain[:])
 _med = median(chain[:])
N_med = median(chain[:N_float])

# Vossler et al. (2022) Wave 1 reference values
p1 = histogram(chain[:], label="", title=" ", density=true)
vline!(p1, [0.217], label="Ref (Vossler et al.)", color=:black, linestyle=:dash, linewidth=2)
vline!(p1, [ _med], label="Posterior median", color=:red, linestyle=:dash, linewidth=2)

p2 = histogram(chain[:], label="", title=" ", density=true)
vline!(p2, [0.197], label="Ref -log(1-  )(Vossler et al.)", color=:black, linestyle=:dash, li
vline!(p2, [ _med], label="Posterior median", color=:red, linestyle=:dash, linewidth=2)

p3 = histogram(chain[:], label="", title=" ", density=true)
vline!(p3, [0.0054], label="Ref (Vossler et al.)", color=:black, linestyle=:dash, linewidth=2
vline!(p3, [ _med], label="Posterior median", color=:red, linestyle=:dash, linewidth=2)

# N is wave-specific
p4 = histogram(chain[:N_float], label="", title="N (Wave 2)", density=true, bins=50)
vline!(p4, [N_med], label="Posterior median", color=:red, linestyle=:dash, linewidth=2)

plot(p1, p2, p3, p4, layout=(2,2), plot_title="Posterior Distributions - Wave 2")
```



Plot trace plots for MCMC diagnostics.

```
plot(chain, plot_title="Trace Plots - Wave 2")
```



Parameter Estimates

We summarise the posterior estimates for the model parameters.

```
 _chain  = vec(chain[: ])
 _chain  = vec(chain[: ])
 _chain  = vec(chain[: ])
N_chain  = vec(chain[:N_float])
```

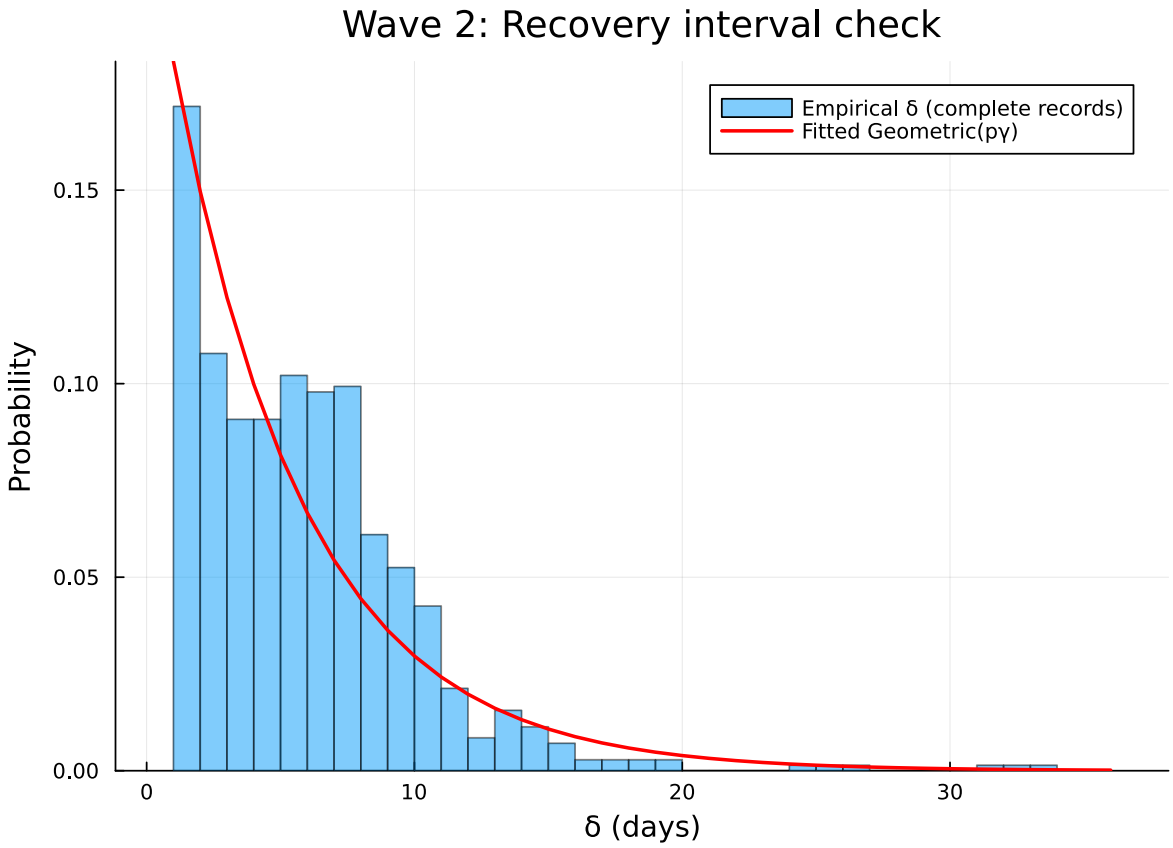
```
p _chain = rate_to_proportion.( _chain)

R0_chain       =  _chain ./ p _chain           # R0 of the discrete process
IP_steps_chain = 1 ./ p _chain                  # mean infectious period in steps (1-indexed G
I0_chain       =  _chain .* N_chain             # implied initial infected count
tau_chain      = K_total ./ N_chain             # observed attack rate posterior

# Posterior point estimates (medians)
 _post  = median( _chain)
 _post  = median( _chain)
 _post  = median( _chain)
N_post  = median(N_chain)
p _post = rate_to_proportion( _post)

function posterior_summary(x; digits=4)
    (median = round(median(x); digits=digits),
     mean   = round(mean(x);   digits=digits),
     q025   = round(quantile(x, 0.025); digits=digits),
     q975   = round(quantile(x, 0.975); digits=digits))
end

posterior_table = (
    N         = posterior_summary(N_chain;        digits=0),
              = posterior_summary( _chain;        digits=4),
              = posterior_summary( _chain;        digits=4),
              = posterior_summary( _chain;        digits=5),
    R0        = posterior_summary(R0_chain;       digits=3),
    IP_steps  = posterior_summary(IP_steps_chain; digits=2),
    I0        = posterior_summary(I0_chain;       digits=2),
    K_over_N  = posterior_summary(tau_chain;      digits=4),
)
```

```
(N = (median = 3926.0, mean = 3982.0, q025 = 3280.0, q975 = 5009.0),   = (median = 0.225, mea

N           median = 3926.0   mean = 3982.0   95% CI = [3280.0, 5009.0]
            median = 0.225   mean = 0.2249   95% CI = [0.2106, 0.2391]
            median = 0.2024   mean = 0.2024   95% CI = [0.1878, 0.2173]
            median = 0.00523   mean = 0.00525   95% CI = [0.00444, 0.00617]
R0          median = 1.228   mean = 1.228   95% CI = [1.182, 1.274]
IP_steps    median = 5.46   mean = 5.46   95% CI = [5.12, 5.84]
I0          median = 20.72   mean = 20.88   95% CI = [16.26, 26.44]
```

```
K_over_N    median = 0.2812   mean = 0.2805   95% CI = [0.2204, 0.3365]

Vossler et al. (2022) Wave 2 reference (continuous-time,  / ):
    = 0.217,   = 0.179,   = 0.0054,  /  = 1.214, 1/  = 5.6 d
```

Recovery interval fit check

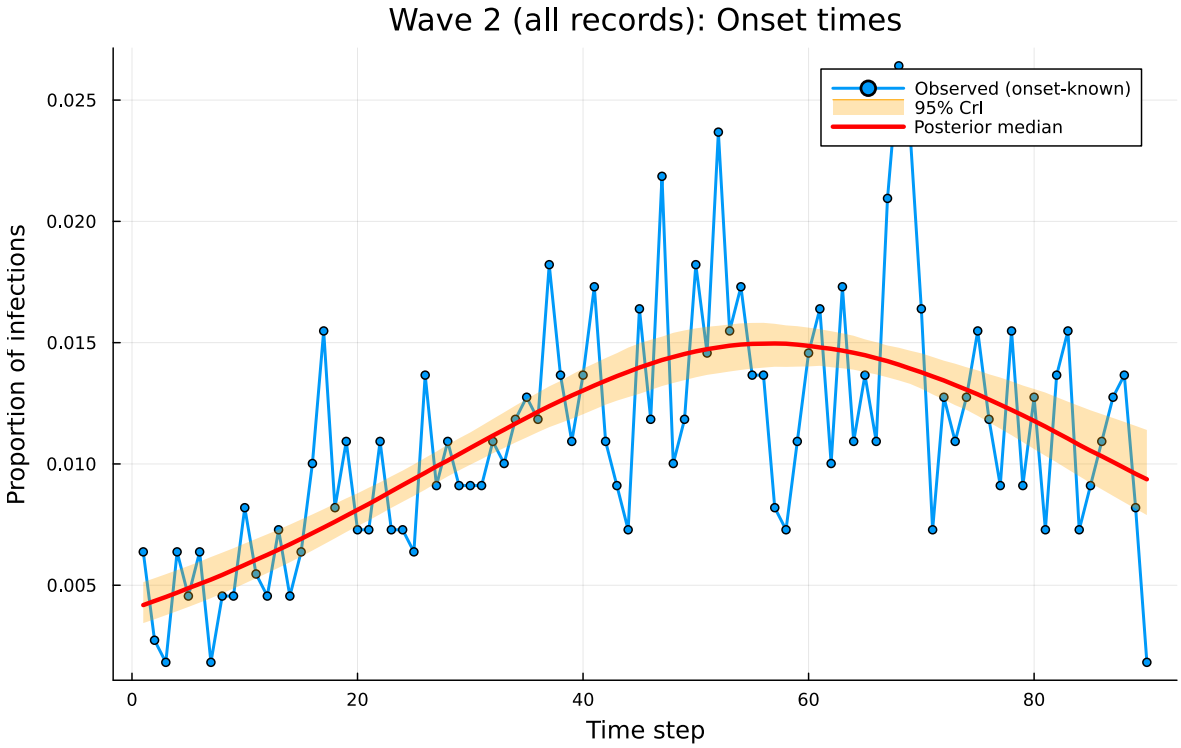
```
# Empirical   histogram vs fitted Geometric(p _post) PMF (1-indexed)
  = delta_complete
kmax = maximum( )
ks = 1:kmax
pmf = pdf.(LocationScale(1, 1, Geometric(p _post)), ks)

p  = histogram( ;
    normalize=true,
    bins=1:kmax,
    alpha=0.5,
    label="Empirical   (complete records)",
    xlabel="  (days)",
    ylabel="Probability",
    title="Wave 2: Recovery interval check"
)
plot!(p , ks, pmf; linewidth=2, color=:red, label="Fitted Geometric(p )")
p
```



Epidemic curve

```
sol,  , f = simulate( _post,  _post,  _post, nsteps)

n_samples_ci = min(500, length(chain[:,1]))
sample_idx   = rand(1:length(chain[:,1]), n_samples_ci)
f_samples    = Matrix{Float64}(undef, nsteps, n_samples_ci)
for (idx, i) in enumerate(sample_idx)
    _, _, f_i = simulate(chain[:,1][i], chain[:,2][i], chain[:,3][i], nsteps)
    f_samples[:, idx] = f_i
end
f_median = [quantile(f_samples[t, :], 0.5)   for t in 1:nsteps]
f_lower  = [quantile(f_samples[t, :], 0.025) for t in 1:nsteps]
f_upper  = [quantile(f_samples[t, :], 0.975) for t in 1:nsteps]

p1 = plot(1:nsteps, th_onset ./ K_onset,
```

```
          label="Observed (onset-known)",
          title="Wave 2 (all records): Onset times",
          xlabel="Time step", ylabel="Proportion of infections",
          linewidth=2, marker=:circle, markersize=3)
plot!(p1, 1:nsteps, f_lower,
      fillrange=f_upper, fillalpha=0.3, fillcolor=:orange,
      linecolor=:orange, label="95% CrI", linewidth=0)
plot!(p1, 1:nsteps, f_median, label="Posterior median", linewidth=3, color=:red)
plot(p1, size=(800, 500))
```



Discussion

Among the three waves, Wave 2 has the largest fraction of onset-only records: 392 of 1,104 (35.5%), compared with 24.5% in Wave 1 (222 of 907) and ~30.0% in Wave 3 (321 of 1,069). Despite this higher missingness in the removal time, the DDSA estimates remain consistent with those reported by Vossler et al. (2022).